

Integration — Lesson 10 Test: Area Between Two Curves

10 questions · show your work in the box · calculators allowed · remember: area = $\int$ (top – bottom) dx

Name: _____ Date: _____ Score: _____ / 10

- 1.** Find the area between the horizontal lines $y = 9$ and $y = 4$ from $x = 0$ to $x = 6$. Check your answer with the rectangle formula (width $\times$ height). 1 pt

- 2.** Find the area between the parallel lines $y = x + 7$ and $y = x + 3$ from $x = 0$ to $x = 5$. 1 pt

- 3.** Find the area between $y = 10$ and $y = 2x$ from $x = 0$ to $x = 4$. (First check which one is on top on this interval.) 1 pt

4. Between $x = 0$ and $x = 1$, which curve is on top: $y = x^2$ or $y = x^3$? Prove your answer by testing the point $x = \frac{1}{2}$. (No integrating needed.) 1 pt

5. Find the area between $y = 9$ and $y = x^2$ from $x = 0$ to $x = 3$. (Notice the parabola meets the line exactly at $x = 3$.) 1 pt

6. Find the area of the region enclosed by $y = 3x$ and $y = x^2$. (No limits given — set the functions equal to find the crossings first.) 1 pt

7. Runner A runs at a steady 7 m/s and runner B at a steady 4 m/s, starting together. Use an area-between-curves integral to find A's lead after 20 seconds. 1 pt

8. Faucet A pours at a steady 8 liters per minute. Faucet B pours at $2t$ liters per minute at time t . Their rates become equal at some time — find that time, then find how many more liters A has delivered than B by that moment. 1 pt

9. A shop earns revenue at a steady rate of 20 dollars per day and spends at a rate of $8 + 2t$ dollars per day at time t . Find the total profit from $t = 0$ to $t = 6$ days. (Profit accumulates at the rate revenue – cost.) 1 pt

10. CHALLENGE: find the area of the region enclosed by $y = x^2$ and $y = 2x + 3$. Show all three steps: crossings, top test, and the integral. 1 pt
